

EMERSON EIMS

BUILDING SUITE PRO

New Project

Location: Not specified
Total Area: 450 m² | Units: 3 | Storeys: 2
Architectural Style: Modern
Date: 26 April 2026

PROJECT INFORMATION	
Project ID	96c9413f-8b7d-4619-aa7d-3d7eedb15ffa
Location	Not specified
Total Area	450 m²
Units / Apartments	3
Storeys	2
Bedrooms per Unit	3
Foundation Type	Strip
Style	Modern
Report Generated	2026-04-26 18:02

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1. PROJECT SUMMARY

Phase phase_1: Site Analysis

water_table_depth	3.5
water_table_source	regional_estimate
water_table_note	Regional estimate from climate/soil model -- verify with on-site geotechnical su
soil_type	clay
vegetation_coverage	0.65
slope_angle	8.25
suitability_pct	90
nearby_structures	survey_required
land_use	residential
climate_zone	tropical
analysis_method	regional_data_model
data_source	EIMS Regional Database + GPS Coordinates

Phase phase_2: Geotechnical Design

bearing_capacity_kPa	150
soil_classification	CH
recommended_foundation	STRIP
settlement_mm	40
slope_stability	stable
water_table_depth	3.5
excavation_depth	2.8

Phase phase_3: Foundation Design

foundation_type	PAD
foundation_depth_m	1.65
footing_width_m	2.48
total_building_load_kN	3375.0
bearing_capacity_kPa	150
safety_factor	3.0
concrete_volume_m3	111.38
reinforcement_steel_kg	10023.8
soil_type	clay
water_table_factor	standard
design_standard	BS 8004 / Eurocode 7
bim_ready	True

Phase phase_4: Floor Plans

floor_count	3
total_area_sqm	450.0
bathrooms	3
rooms	{'bedrooms': 4, 'bathrooms': 3, 'living_areas': 1, 'kitchen': 1, 'extra_rooms':

style	modern
area_per_unit	450.0
building_code_compliance	True
area_efficiency_percent	80.0
cost_per_sqm_estimate	calculated_in_costing_phase

Phase phase_5: Electrical Design

total_connected_load_kw	9.38
max_demand_kw	5.63
diversity_factor	0.6
supply_type	single-phase 230V
main_current_A	24.5
lighting_points	12
socket_outlets	12
total_circuits	7
lighting_circuits	2
socket_circuits	2
fixed_appliance_circuits	3
distribution_boards	1

Phase phase_6: Plumbing Design

total_occupants	6
daily_demand_liters	900
peak_hourly_liters	90
total_fixture_units	34.5
main_supply_pipe_mm	32
cold_water_pipes	11
hot_water_pipes	7
waste_outlets	10
soil_stacks	1
hot_water_storage_liters	240
water_heater_kw	9.6
storage_tank_liters	1000

Phase phase_7: BOQ

material_items	77
total_material_cost	103127.21
currency	USD
items_by_category	{'external': 12, 'electrical': 11, 'plumbing': 14, 'structural': 17, 'mep_hvac':
items	[{'item_no': 1, 'description': 'Portland Cement (50kg)', 'unit': 'bag', 'quantit
calculation_basis	{'total_area_m2': 450.0, 'stories': 3, 'units': 1, 'bedrooms_per_unit': 4, 'foot
waste_factor_applied	True
method	quantity_takeoff_from_dimensions

Phase phase_8: Infrastructure

solar_system	{'panels_kw': 9.0, 'panels_count': 23, 'battery_kWh': 13.5, 'estimated_cost_usd': 1676.88}
borehole_system	{'depth_meters': 8, 'estimated_yield_lpm': 15, 'pump_kw': 3, 'estimated_cost_usd': 1676.88}
grid_option	{'connection_cost_usd': 375, 'monthly_tariff_estimate_usd': 67}
recommendation	solar_grid_hybrid
estimated_annual_saving_usd	1676.88

Phase phase_9: Landscape Design

lawn_area_m2	157.5
hardscape_area_m2	45.0
garden_area_m2	67.5
plants_count	540
irrigation_system	True
recommended_style	tropical_lush
estimated_cost_usd	1552
maintenance_monthly_usd	31

Phase phase_10: Permits & Compliance

jurisdiction	Spain
building_code	CTE 2019
building_code_compliance	REQUIRES_VERIFICATION
permits_required	6
estimated_timeline_days	45
estimated_fees_usd	1080
setback_compliance	{'front_m': 5, 'side_m': 3, 'rear_m': 3}
required_documents	['Architectural drawings (stamped)', 'Structural calculations (engineer-certified)']

Phase phase_11: 3D Visualization

status	ready
model_endpoint	/api/drawings/3d-model

Phase phase_12: Live Costing

material_cost	103127.21
labor_cost	32485.07
equipment_cost	8250.18
overhead	16273.47
contingency	19216.31
subtotal	179352.24
contractor_profit	35870.45
total_project_cost	215222.69
cost_per_sqm	478.27
labor_multiplier	0.7
region	Spain
currency	USD

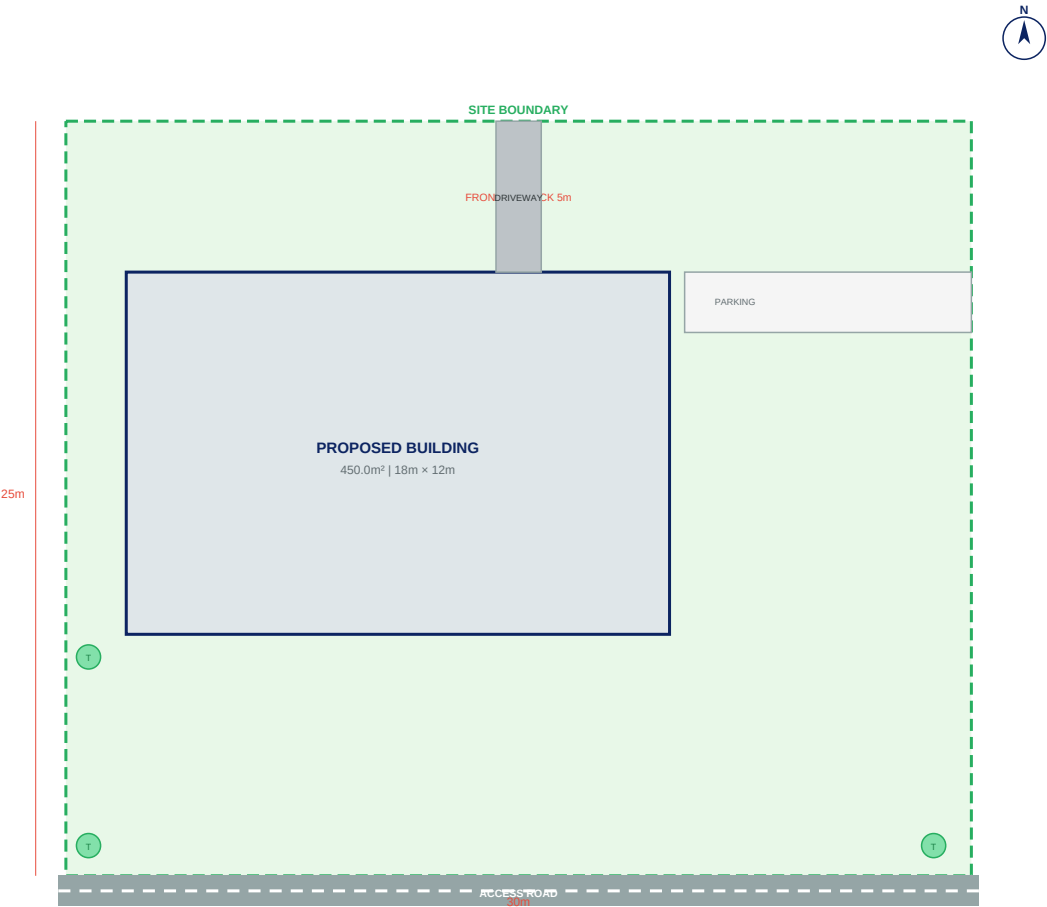
Phase phase_13: Integration Layer

status	complete
export_formats	['IFC2x3', 'FBX', 'DXF', 'PDF', 'XLSX']

2. SITE PLAN

Site layout showing building footprint, access roads, setbacks and landscaping. Area: 450 m²

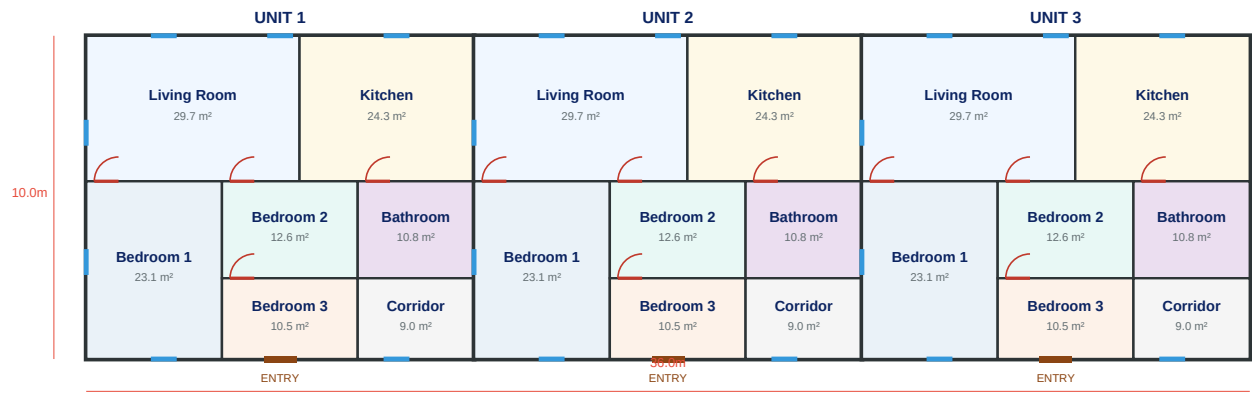
SITE PLAN - Not specified | Plot: 30m×25m



3. ARCHITECTURAL FLOOR PLAN

Ground floor layout — 3 unit(s), 3 bedrooms, showing room dimensions, doors, windows and circulation.

FLOOR PLAN - 450.0m² | 3 Units | 2 Storey(s)

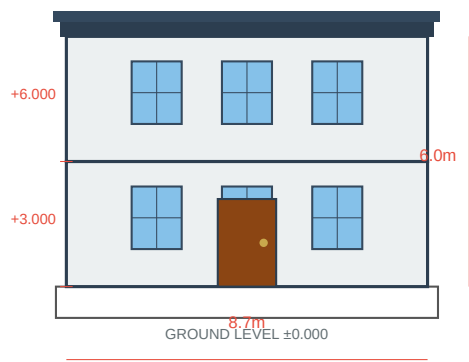


4. BUILDING ELEVATIONS

4a. NORTH ELEVATION

Front view showing 2 storey(s), window/door openings, roof profile and floor levels.

NORTH ELEVATION - 2 Stories | Style: Modern

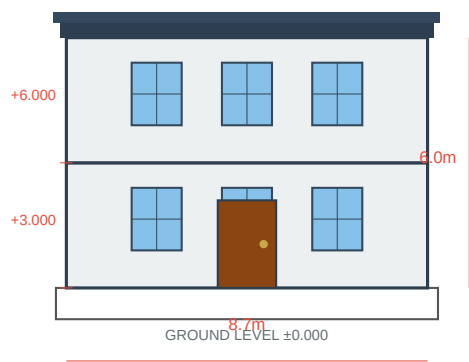


EMERSON EIMS | NORTH ELEVATION Stories: 2 | Height: 6.0m
Scale 1:50 | 2026-04-26 | Drawing: EL-N01 Style: Modern | IFC Ready

4b. EAST ELEVATION

Side view showing building depth, roof slope and storey heights.

EAST ELEVATION - 2 Stories | Style: Modern

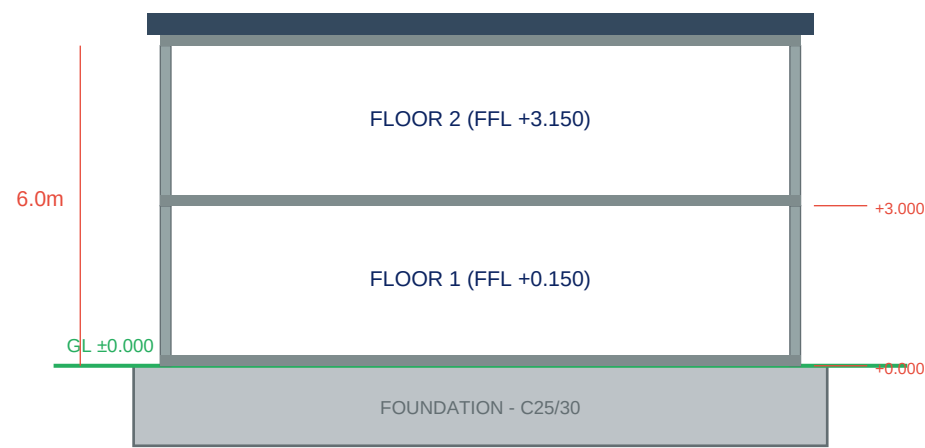


EMERSON EIMS | EAST ELEVATION Stories: 2 | Height: 6.0m
Scale 1:50 | 2026-04-26 | Drawing: EL-E01 Style: Modern | IFC Ready

5. BUILDING CROSS SECTION

Vertical cut through building showing floor slabs, storey heights (2 floors), foundation depth, roof structure and structural members.

CROSS SECTION A-A | 2 Stories



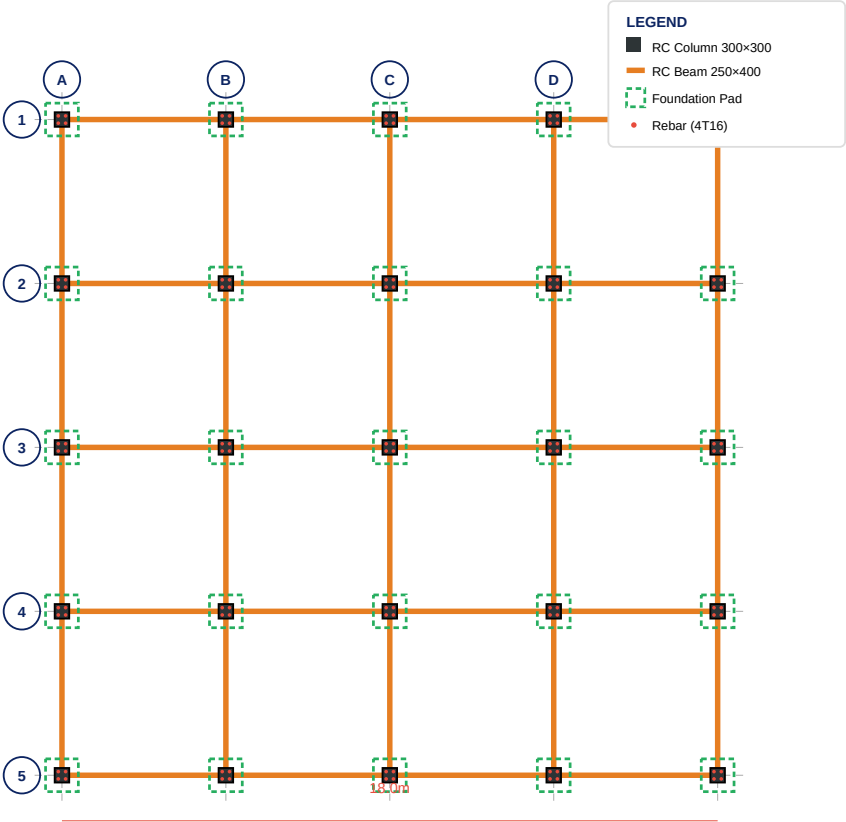
EMERSON EIMS | BUILDING SECTION A-A

2026-04-26 | SEC-001 | Scale 1:50

6. STRUCTURAL LAYOUT

Column grid, beam layout and foundation pads. Foundation type: Strip. All dimensions in metres. Concrete grade C25/30, Steel Y16.

STRUCTURAL LAYOUT - Strip Foundation | 2F | Grid 4.5m×4.5m



EMERSON EIMS | STRUCTURAL LAYOUT

Grid: 4.5m x 4.5m | Columns: 25 | 2026-04-26 | STR-001

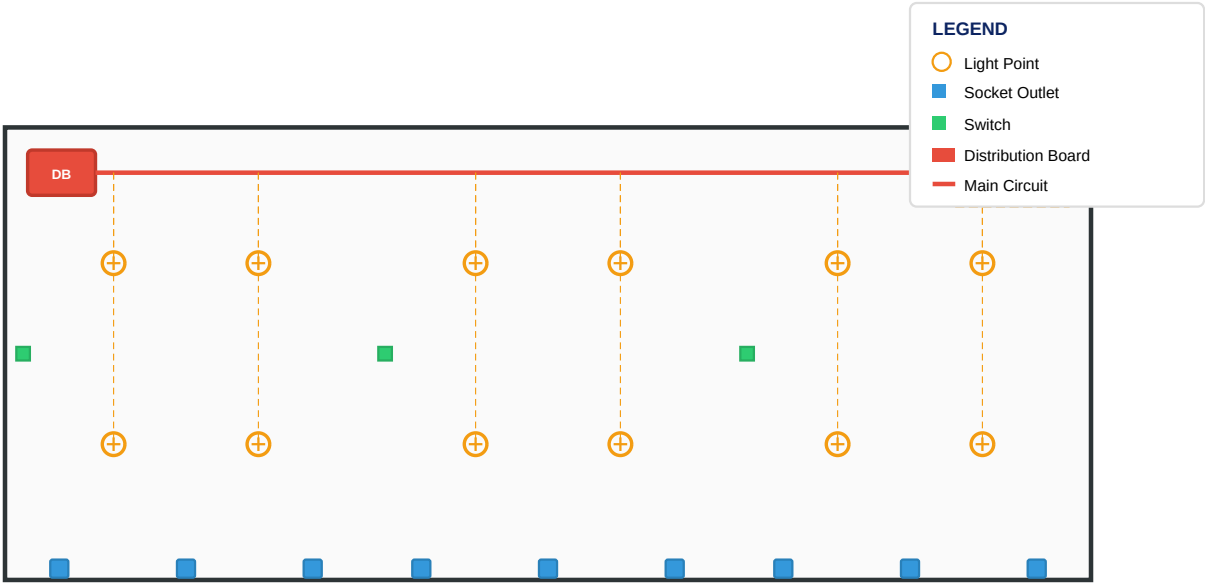
Concrete: C25/30 | Steel: Y16

Foundation: Strip | FOS ≥ 3.0

7. MEP LAYOUT — ELECTRICAL

Electrical distribution layout showing DB positions, circuit runs, socket/light points and earthing.

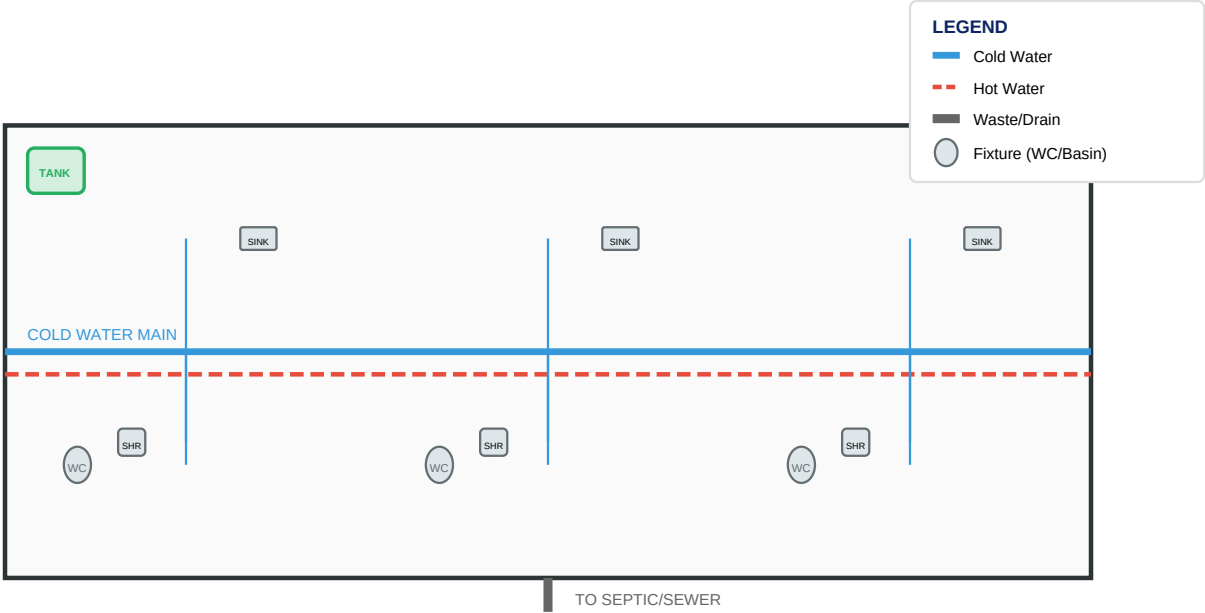
ELECTRICAL LAYOUT - 450.0m² | 3 Units



8. MEP LAYOUT — PLUMBING

Plumbing layout showing supply/drain runs, fixture locations, water heater and main shutoff valve.

PLUMBING LAYOUT - 450.0m² | 3 Units



9. COST SUMMARY

Cost data not yet generated. Run the analysis first.

This report was generated by EMERSON EIMS Building Suite Pro. All drawings are indicative and must be verified by a licensed engineer before construction.